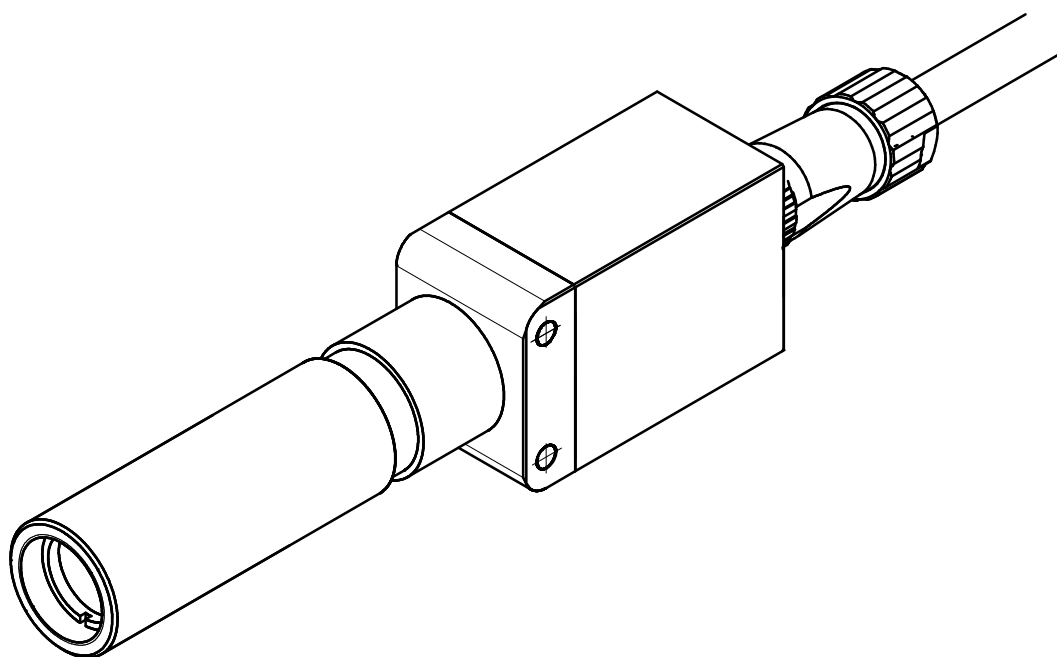


Instruction Manual

INFRARED THERMOMETER

F L H X Series

Laser sight type



JAPANESE CORPORATION

2nd Edition

CAUTION



1. When not keeping the following matters, a temperature indicating value has been displayed lower than the actual temperature.
Especially pay attention in case of controlling temperature by output of thermometer because it leads to burnout accidents by over doing of the heating device.
 - 1) Don' t interrupt the light path even if a part between the lens part of the thermometer and work piece.
 - 2) Check that there is not dirt in the lens part of the thermometer daily.
When it is dirty, clean it with alcohol.
 - 3) Check that the sight suits a point for the purpose surely daily.
When sifted, redo a sight and fix a sensor head firmly.



2. In case of the different work piece, emissivity is also often different each.
Measure the temperature with setting emissivity of each work piece.



3. Proofread at the frequency once a year by the blackbody-furnace for the thermometer to maintain precision.



4. This unit is a precision instrument. Do not open to the absolute. Once you open and there is a risk of dropping the optical performance and waterproof function, can not be returned to the original.



5. The thermometer, applications such as related to human life (the operator of body temperature check, management and control of the working chamber, etc.) are not taken into account. Please stop use.



6. The thermometer, high-risk (medical, automotive, nuclear, military, space equipment, etc.) are currently not considered for use to. Please stop use for applications that could cause serious damage.



Warning

Do not stare into the sighting laser light emanating from the sensor head tip.
Also that you do not see the direct beam with optical instruments.
There is a risk to cause damage to your eyes.

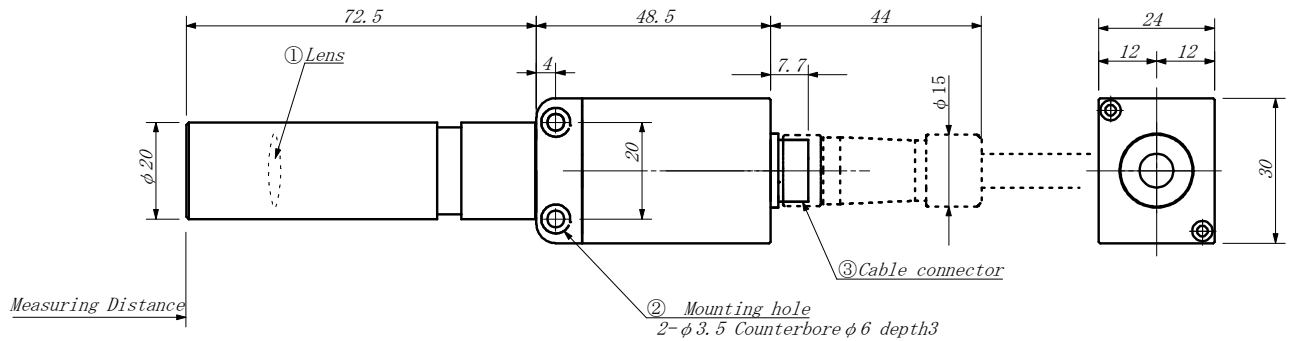
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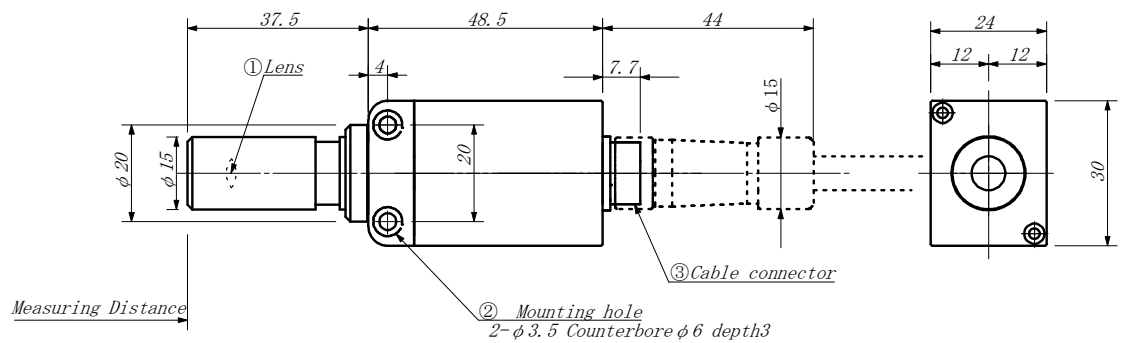
Name of Each Part and Installation

Sensor Head

φ 15 Head



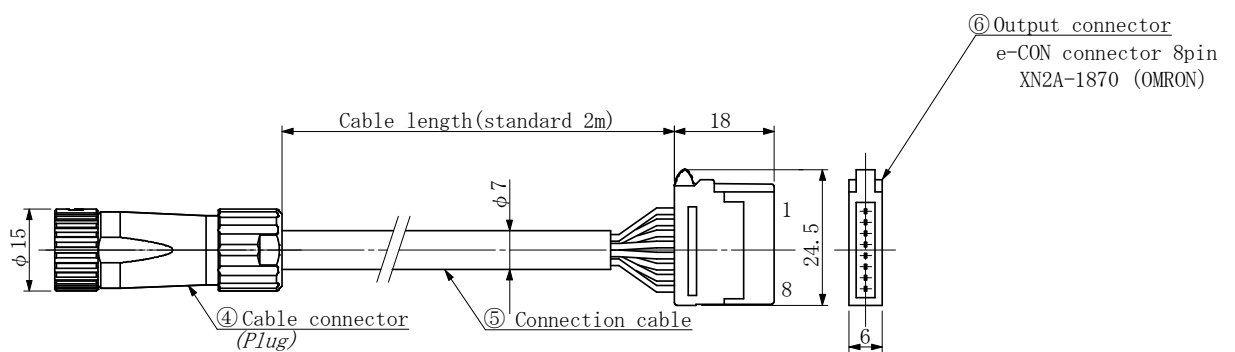
φ 6 Head



(Unit:mm)

- ① Lens :It condenses infrared rays from work piece.
- ② Mounting hole :Through hole for fixing the sensor head. (φ3.5)
- ③ Cable connector :It connects a temperature converter and connection cable. (Receptacle)

Accessory : Connection Cable



(Unit:mm)

- ④ Cable connector : It connects a temperature converter and connection cable. (Plug)
- ⑤ Connection cable : It connects the power and the temperature output.
- ⑥ Output connector : It connects to external instruments. e-CON 8P

Installation and Connection

[Fixation of Sensor head]

Adjusts its direction to the work piece in sensor head lens part①, and set the distance between the sensor head tip and the work piece to be situated to the measuring distance of the specification.



Caution

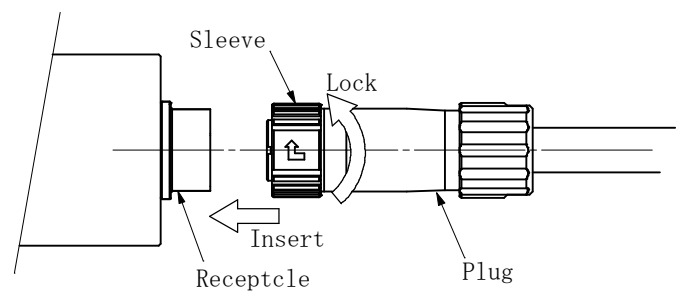
This unit is a precision instrument. When mounting, mounting to the thermometer, such as stress is applied (strongly cable And put placement, excessive bending, be sure to strongly etc.) and sandwich, is unreasonable placement avoid

[Connection of Cable] Bayonet lock type

- 1) A keyway of a cable connector(plug)④ of connection cable⑤ and a key of a cable connector(receptacle)③ of sensor head are put in and plug is inserted in a receptacle.

After inserting a plug to the inside, a sleeve is turned to the way of the arrow and it's locked. Even if it's inserted, it isn't locked automatically, so please be sure to do rock movement.

- 2) When removing a connection, I make the left revolve with a sleeve of a plug. Locking is contrary, and it's possible to remove.

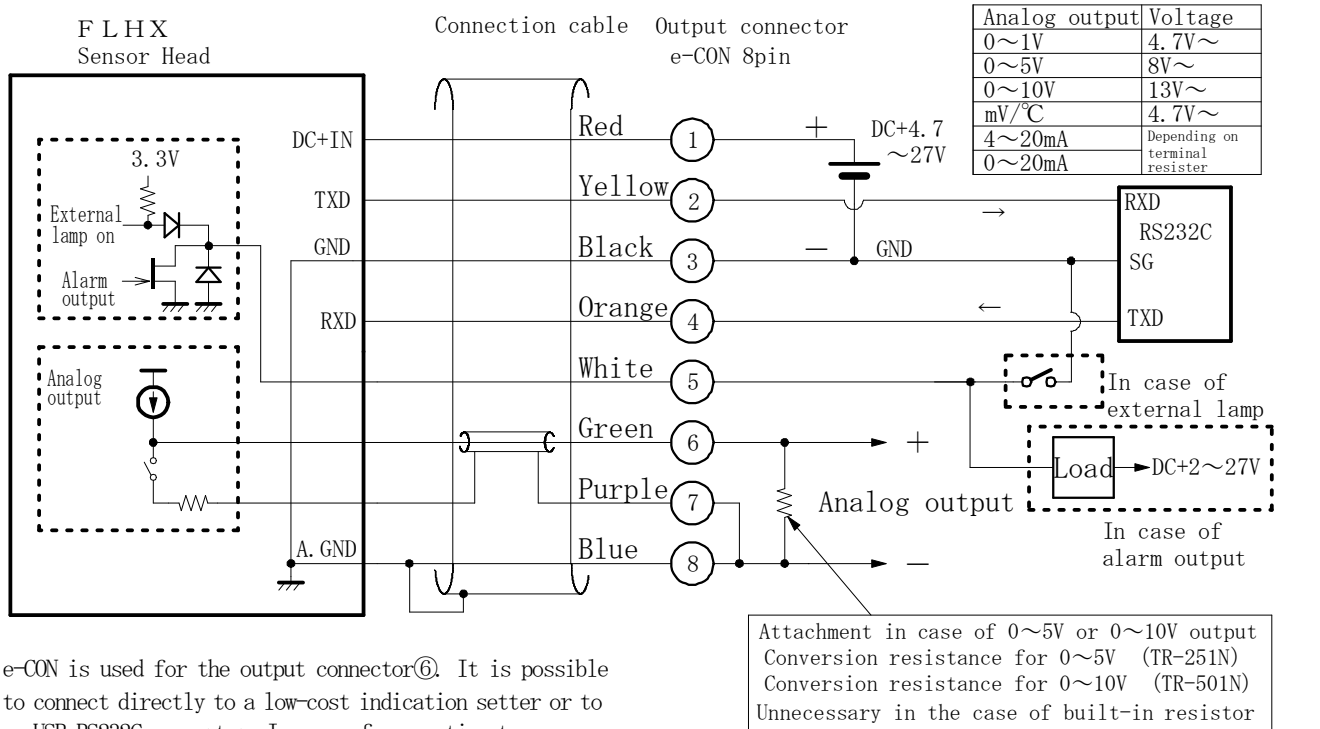


Caution

Connecting or disconnecting of the instrument should be performed in the state always turn off the power. Cause a failure may cause.

Wiring

Connect a connection Cable as below.



e-CON is used for the output connector⑥. It is possible to connect directly to a low-cost indication setter or to an USB-RS232C converter. In case of connecting to a conversion terminal or etc., use it after removing the lead wire of connection cable⑤ from e-CON.

- (1) By using a slotted driver, put it into the vent of the e-CON until its operation lever(white) is locked, and pull out the lead wire.
- (2) Put a slotted driver into cancellation vent, pull the lever. So the operation lever is returned with the snap. After having pulled out the lead wire, let it return by all means.

[Power Supply]

Connect DC power supply (DC+4.7 ~27V) to No. 1+, No. 3- of output Connector⑥.

[Analog Output]

Analog is output from No. 6+, No. 7, 8- of output connector⑥.

As for No. 7 and No. 8, short at its terminal side.

In case of need for 0~5V or 0~10V output, attach an optional conversion resistance (TR-251N or TR-501N) in parallel against analog output. Switch its analog output to 0~20mA. (Unnecessary in the case of built-in resistor)

[RS232C Output]

RS232 is output from No. 2, 3, 4 of output connector⑥.

GND connects to No. 3 shared with DC power supply.

[Ex-Lighting / Alarm Output]

It is possible to use either external lighting or alarm output.

It is chosen by parameter setting using RS232C.

- External lighting Connect the switch for light on/off between No. 5 and No. 3 of output connector⑥.
- Alarm output Connect the load to No. 5 of output connector⑥.
The power supply minus for the load is connected to No. 3.



Caution

As false connection (reverse connection of power supply) and the short circuit of the output cause the trouble, please be careful.

Parameter Setting

To change the parameter setting of the thermometer, use the parameter setter PWCX, the indicating setter TMCX or the parameter setting software PWSX. At this instruction manual, it is explained to change its setting by using a PWCX. For detailed usage of each parameter, see a PWCX instruction manual.

Communication Condition

Setting communication parameter of RS232C is the following.

Communication Max Speed	Coñ5	[MODE] key more than 3 seconds → [MODE] key × 7
Communication Initial Speed	bP5	above → [ENT] key × 1
Parity bit	PAR1	above → [ENT] key × 1
Stop bit	StoP	above → [ENT] key × 1
Data length	dLEn	above → [ENT] key × 1

Sight

[Laser Sight]

Sight lamp ON/OFF	LAAp	[MODE] key × 1 → Select on by ▼ ▲ key and [ENT] key × 1
-------------------	-------------	--

Adjust and install its sensor head so that its sight light suits measuring position of the work piece.

It is easy to use an optional mounting bracket of the head (TMAX-A).

Push [MODE] key × 6 and return it to the measurement mode, and it makes sight light off.



Caution

Set spot size to be smaller than the work piece.

If it is set larger than it, its temperature is indicated lower.

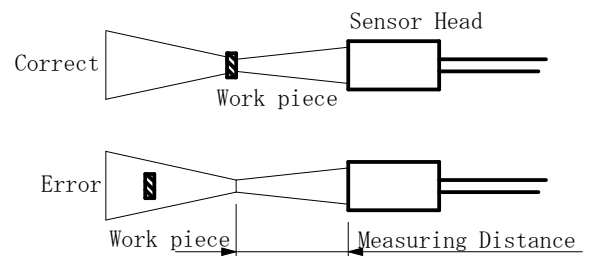


Warning

Do not stare into the sighting laser light emanating from the sensor head tip.

Also that you do not see the direct beam with optical instruments.

There is a risk to cause damage to your eyes.



As factory default, “Sight Lamp Control” is set in regular as “ON”. “Sight lamp ON/OFF” does not function.

Sight Lamp Control	LAAc	[MODE] key more than 3 seconds → [ENT] key × 1
--------------------	-------------	--

[External Lighting Input]

“Sight lamp ON/OFF” is also controlled by No.5 terminal of a thermometer output connector.

Ext pin Function	Alt	[MODE] key more than 3 seconds → [ENT] key × 8
	Select Eyt by ▼ ▲ key and [ENT] key × 1	

If the No.5 terminal output connector is shorted-circuit with GND, the lamp turns on in spite of its setting of “Sight Lamp Control”.

Temperature Measurement

1. Supply a power supply to the sensor head. (Refer to P.4 Wiring)
2. Set the emissivity and the reflection value of the work piece as needed.
Setting of emissivity (refer to P.6 Measurement of Emissivity and P.7 Emissivity Setting)
Setting of reflection value (Refer to P.7 Reflection Correction)
3. Measured temperature is output or indicated. (Refer to P.8 Temperature Output)



Caution

- ① Emissivity value depends on the surface condition and so on of the work piece.
In case of the black body, it is 1.0. As for the general object because it is smaller than 1.0, their infrared ray is radiated lower than that of black body.
Therefore, for exact temperature measurement, it is necessary to set and to correct emissivity of the work piece beforehand. (P.18 Emissivity Reference Data)
- ② Use it in ambient temperature of the specification range by all means.
- ③ Because the temperature is indicated higher by reflections of a tungsten lamp, a hot heat source of sunlight near the work piece, take measures such as by shadowing.
This phenomenon becomes the more remarkable at the lower temperature.
In addition, it is better to use a fluorescent lamp or the mercury lamp for the lighting.
- ④ Generally, in case there is dust, gas and moisture, the indicating value declines.
→ The air purge is needed by the optional air purge hood or air controller.
- ⑤ When the lens has been dirty, the temperature indicates lower. In this case, wipe out dirt of the lens regularly by the swab dipped in alcohol. Be careful because a lens is damaged it is rubbed too strong.
→ Protect the lens by the optional air purge hood, the air less hood and the window material.

Measurement of Emissivity

Set it as below at the time of measuring temperature with the infrared thermometer because it depends on its emissivity.



Caution

Set its temperature of the work piece in condition to be high temperature if it is possible. It is not possible to set it less than at 50°C as a standard.

1. The way by the contact type thermometer
 - 1) Measure the temperature of the work piece by the contact type thermometer such as the thermocouple or the thermistor.
In this case be careful enough to measure because temperature is often indicated lower by contact pressure or by contact heat conduction.
 - 2) Mount the sensor head and the work piece almost vertically for decreasing influences of ambient light.
 - 3) After setting its sight by the same way as "Temperature Measurement", set the emissivity value so that its measurement temperature output fits the temperature of the work piece which is measured.
 - 4) This emissivity value is that of the work piece at that temperature.
2. The way by painting the black body
 - 1) Paint with the black body paint whose emissivity is known, to a part of the work piece at more than twice of area of measurement spot. Pay attention because its surface temperature is getting lower if it is painted too thick.
 - 2) After setting sight to part painted with the blackbody by the same way as temperature measurement. Set its emissivity value as that of blackbody painting.
It starts to output measuring temperature which is equal to that of work piece.
 - 3) Set sight to the part not painted with blackbody painting but near that painted.
And set its emissivity as the same temperature as 2).
 - 4) This emissivity value is that of work piece at that temperature.

Emissivity Setting

Emissivity can be set by 3 different ways.

[Value Setting] It sets the emissivity value.

Emissivity(Value setting) **E A 1 5** **MODE**key×2

[Manual Setting] Emissivity is set as confirming temperature indication.

Emissivity(Manual setting) **E A 1 5** **MODE**key×2→**ENT**key×1

[Auto Setting] Emissivity is set automatically by inputting the target temperature.

Emissivity(Auto setting) **E A 1 5** **MODE**key×2→**ENT**key×2



Caution

Its auto setting functions in case the temperature is more than 50°C.
When the auto setting has been unable, it replies as “**RtEr**”.

Reflection Correction

This thermometer has a function to correct the two kinds of reflective elements.

[Ambient Reflection Function]

Ambient Reflection On/Off **A A b** **MODE**key more than 3 seconds→**MODE**key×1→**ENT**key×4

When it is “ON”, it corrects the elements that infrared radiation of the room temperature reflects on the surface of the work piece. In case of measuring outdoor, switch to “OFF”.

[Setting Reflection Value]

It corrects the elements that infrared radiation of high temperature objects reflects on the surface of work piece.
It can be set by 3 different ways such as that of emissivity

Reflection Value(Setting value) **r E F** **MODE**key more than 3 seconds→**MODE**key×1→**ENT**key×5

Reflection Value(Manual setting) **r E F** **MODE**key more than 3 seconds→**MODE**key×1→**ENT**key×6

Reflection Value(Auto setting) **r E F** **MODE**key more than 3 seconds→**MODE**key×1→**ENT**key×7

When the auto setting has been unable, it replies as “**RtEr**”.

Temperature Output

[RS232C Output]

The measured temperature is output from No. 2, 3 and 4 terminals by RS232 signal. (Refer to “Wiring” P.4)

If you are connected to the display setter TMCX, the measured temperature is digitally displayed.

[Analog Output]

The measured temperature is output from No.6 terminal by analog signal. (Refer to “Wiring” P.4)

In case of being necessary of 0~5V or 0~10V, connect an optional resistance(TR-251N or TR-501N) to the analog terminal with parallel and switch the analog output to 0~20mA. (Unnecessary in the case of built-in resistor)

● Analog output1 is the standard equipment and is used by selecting one of below.

Analog output1 type **An 1L** **MODE**key more than 3 seconds→**MODE**key×5
 Select output type by **▼****▲**key and **ENT**key×1

Nothing	nc	
4 ~ 20 mA	4-20	(Factory default)
0 ~ 20 mA	0-20	option external resistor 0-5V/0-10V
0 ~ 1 V	0-1	option built-in resistor 0-5V/0-10V
mV/°C	mV-c	



Caution

The minus temperature is not able to output at mV/°C. It is limited at 0mV.

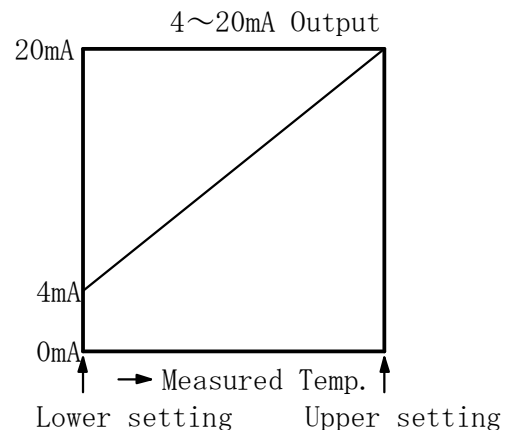
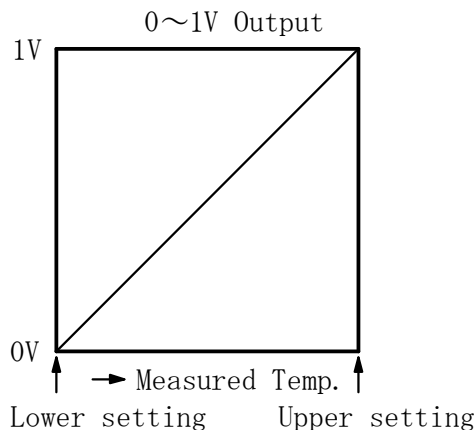
In case of 0~1V or mV/°C output, the output value less than 30mV is not guaranteed.

In the case of 0-5V/0-10V specifications, please refer to the section of the analog output of P.12 specifications.

● In case of 0~20mA, 4~20mA and 0~1V, it is possible to set the scaling setting.

Analog output 1H **An 1H** **MODE**key more than 3 seconds→**MODE**key×5→**ENT**key×1
 Analog output 1L **An 1L** **MODE**key more than 3 seconds→**MODE**key×5→**ENT**key×2

It is also able to be reversed polarity if the value of the setting analog output 1H is set less than that of analog output 1L.



Alarm Output

The alarm signal is output from NO.5 terminal of output connector.

The alarm is selected by Ext pin Function. Alarm is set as factory default.

External pin Function **ALt** **MODE** key more than 3 seconds → **ENT** key × 8
 Select **ALā** by **▼▲** key and **ENT** key × 1

Set alarm operating mode.

Alarm mode 1 **ANd.1** **MODE** key more than 3 seconds → **MODE** key × 2 → **ENT** key × 3

Alarm output operation

Mode No.	Mode name	Measured temp	
		Alarm L set value	Alarm H set value
1	Upper limit ON		
2	Upper limit OFF		
3	Lower limit ON		
4	Lower limit OFF		
5	Upper/Lower limit ON		
6	Upper/Lower limit OFF		
7	Error ON		
8	Error OFF		

Error : internal voltage abnormality.

Mode No7: Error ON, Non-error OFF

Mode No8: Error OFF, Non-error ON

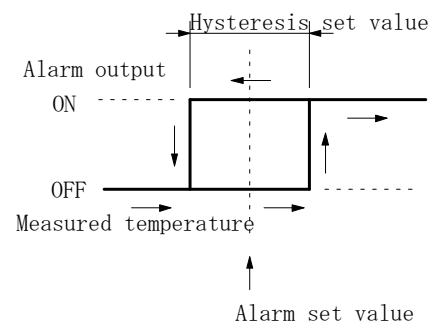
It sets alarm H or alarm L setting value.

Alarm 1H **ALā.H** **MODE** key × 3
 Alarm 1L **ALā.L** above key → **ENT** key × 1

It sets the alarm hysteresis.

Hysteresis **HYSL** **MODE** key more than 3 seconds
 → **MODE** key × 2 → **ENT** key × 2

Alarm output hysteresis operation
 In case of Mode 1 (Upper limit ON)



It selects the signal for the alarm judgment.

Judgment change **JUDE** **MODE** key more than 3 seconds → **MODE** key × 2 → **ENT** key × 4
REAL Real
Hold Hold

Peak Hold

In case that the work piece moves intermittently, it holds the peak temperature of it.
In case of using peak hold, set on "Peak hold ON/OFF".

Peak hold ON/OFF **P.HLD** MODEkey×4
→Select **on** by ▼▲key and ENTkey×1

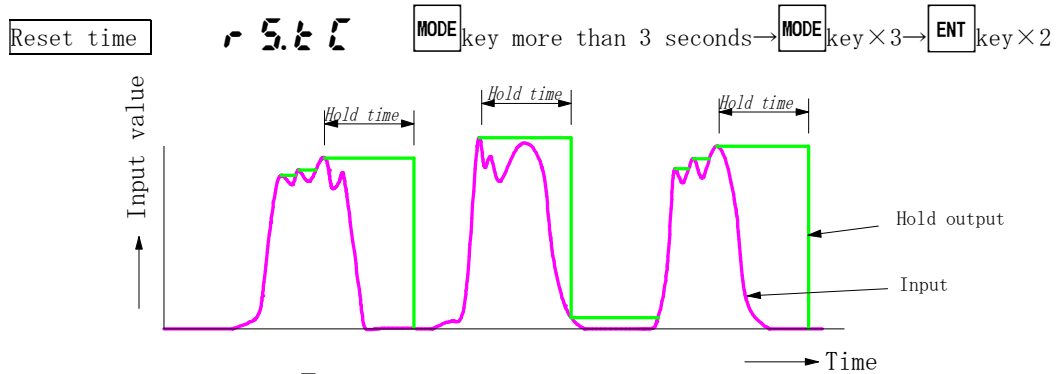
The reset type of the hold is selected from two kinds of the following.

Reset Type **r.S.tP** MODEkey more than 3 seconds→MODEkey×3→ENTkey×1
Select the reset type by ▼▲key and ENTkey×1

●Time Reset **t1AE**

After detecting the peak value, it is held at fixed intervals.

→Set its hold time by peak hold time setting.

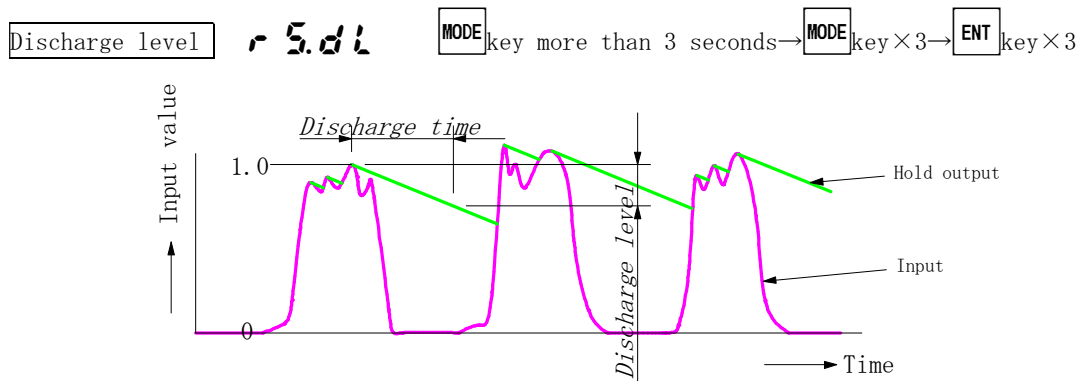


●Discharge Reset **d15**

After detecting the peak value, its level is declined with fixed angle.

It is comparatively stable signal without the large temperature declining at reset.

The angle of discharge is set by peak hold reset setting and by discharge level setting.



Hold output **HLd.o** MODEkey more than 3 seconds→MODEkey×3→ENTkey×4

It is possible to select which signal the peak hold sets.

d15P Only temperature indicating value.

d-R1 Temperature indicating value and analog output1

(Only analog output type and indicating setter are valid.)

ALL ALL (Only indicating setter with analog output 2 is valid.)

Note) Temperature indicating value: The temperature indication of parameter setter PWCX or of indicating setter TMCX.

Smoothing

When it is difficult to read the large change of the measured temperature, it is possible to smooth the moving average.

Smoothing

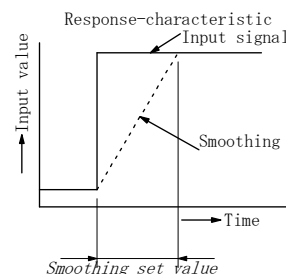
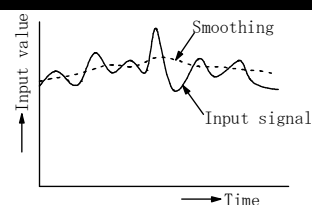
5.000

MODE key × 6

Setting value is small: The faster its response speed is, the larger its change of temperature indicating is.

Setting value is large: The slower its response speed is, the smaller its change of temperature indicating is.

It is set at 0.05s as factory default.



Sensor Correction

It is set in case of correcting the temperature measurement value.

Sensor correction span

SE.SP

MODE key more than 3 seconds → MODE key × 6

It is set in case of the sensitivity correction or of the transmittance correction using window materials.

It is set in case of the correction at high temperature measuring.

At the setting the emissivity, when the temperature indicating value is set with that of the contact type thermometer, in case of the higher temperature indicating value in spite of the setting emissivity at 1.0, it is possible to set the temperature indicating value by the sensor correction span setting value being less than 1.0.

In addition, because the sensor correction span depends on the radiation energy level, change the setting value little by little confirming temperature indicating value.

The temperature is proofread at 1.000 on factory default.



Caution

① When the correction value is set less than 1.000, the upper limit of measuring range declines.

At 1.000, the maximum temperature is indicated 10~50°C higher than that of the specification.

At 0.900, the maximum temperature is as same as that of the specification.

At less than 0.900, the maximum temperature is limited lower than that of the specification.

② Use it at 1.000 generally.

Sensor correction zero

SE.0F

MODE key more than 3 seconds → MODE key × 6 → ENT key × 1

It is set in case of correction of measurement for lower temperature.

It is translated in parallel to all the measuring range. Also it supports a temperature indicating value linearly.

It is proofread at 0 as factory default.

Initialization Data Reading

It returns each parameter to the setting value as factory default.

It is used for the restoration to original condition in case of the fault value being set.

Initialization

i n i . r

MODE key more than 3 seconds → ENT key × 6 → ENT key × 1

→ select ON by ▼/▲ key and ENT key × 1

Specification

Temperature Converter Specification

Model	F L H X - A N N		F L H X - P N N	F L H X - T N N
Measuring range				
ϕ 0.6	5 0 0 ~ 2 0 0 0 °C		2 2 0 ~ 1 6 5 0 °C	9 0 ~ 1 5 0 0 °C
ϕ 0.2	6 0 0 ~ 2 0 0 0 °C		3 0 0 ~ 2 0 0 0 °C	1 6 0 ~ 2 0 0 0 °C
ϕ 0.1	7 0 0 ~ 2 0 0 0 °C		4 0 0 ~ 2 0 0 0 °C	2 2 0 ~ 2 0 0 0 °C
Effective wavelength	0 . 8 ~ 1 . 0 μ m		0 . 8 ~ 1 . 6 μ m	1 . 9 5 ~ 2 . 6 μ m
Detective element	S i		I n G a A s	I n G a A s
Sight	Laser sight (class 1)			
Accuracy *note1	Less than 8 0 0 °C : ± 4 °C 8 0 0 ~ 1 2 0 0 °C : Measured value $\pm$ 0 . 5 % 1 2 0 0 ~ 2 0 0 0 °C : Measured value $\pm$ 1 . 0 %			
Repeatability *note1	Measured value $\pm$ 0 . 2 % $\pm$ 2 °C			
Measuring resolution *note1	Lowest temperature+Less than 5 0 °C : 3 °C or Less Lowest temperature+Less than 1 0 0 °C : 1 °C or Less Lowest temperature+More than 1 0 0 °C : 0 . 5 °C or Less			
Response time	0 . 0 0 1 ~ 5 s (0~95% analog output) Changeable by smoothing function (Range switching is 0.002 s are added)			
Output	Analog Output(Non-isolation) 0 ~ 2 0 m A / 4 ~ 2 0 m A / 0 ~ 1 V / m V / °C select Option 0 ~ 5 V / 0 ~ 1 0 V Output type(Changeable) Output valid range Output accuracy*Note 0 ~ 2 0 m A More than 0 . 2 m A $\pm$ 0 . 0 2 m A 4 ~ 2 0 m A More than 4 m A $\pm$ 0 . 0 2 m A 0 ~ 1 V spec More than 3 0 m V $\pm$ 1 . 5 m V m V / °C More than 3 0 m V (3 0 °C) $\pm$ 1 . 5 m V 0 ~ 5 V spec More than 5 0 m V $\pm$ 5 m V m V / °C More than 5 0 m V (5 0 °C) $\pm$ 5 m V 0 ~ 1 0 V spec More than 1 0 0 m V $\pm$ 1 0 m V m V / °C More than 1 0 0 m V (1 0 0 °C) $\pm$ 1 0 m V *Note: Above output accuracy(static error) is added to the accuracy(temperature). R S 2 3 2 C Output(R S 2 3 2 C conformity, Non-isolation) Output swing range : about $\pm$ 4 V Baudrate : 4800, 9600, 19200, 38400, 57600, 115200bps Alarm output(Non-isolated) One open drain output:D C 2 7 V , 0 . 2 A Hysteresis: 0 ~ 9 9 . 9 °C			
Peak hold	Reset method (selection)	Time : 0 . 0 1 ~ 1 0 s changeable Discharge : Time 0 . 0 1 ~ 1 0 s 、 Level 0 . 2 ~ 1 . 0		
Emissivity correction	Guarantee range : 0 . 3 ~ 1 . 0 Setting range : 0 . 0 5 0 ~ 1 . 0 0 0 (Setting resolution 0 . 0 0 1)			
Sensor correction function	Span : 0 . 5 0 0 ~ 2 . 0 0 0 Zero : - 5 0 ~ + 5 0 °C (° F)			
Temperature indication	Nothing			
Warm up time	1 min			
Structure	Dust proof (I P 6 7 equivalency) , without the output connector part			
Operating ambient temp.	0 ~ 5 0 °C			
Operating ambient hum.	3 0 ~ 8 5 % R H (without dew drop)			
Storage temperature	- 1 5 ~ 7 0 °C			
Supply voltage	D C 4 . 7 ~ 2 7 V 、 0 . 1 A m a x			
Weight	ϕ 1 5 Head : 1 0 5 g 、 ϕ 6 Head : 8 0 g (without cables)			

*Notel. Ambient temperature 23 ± 5 °C, Emissivity 1.0, Smoothing time 0.05s.

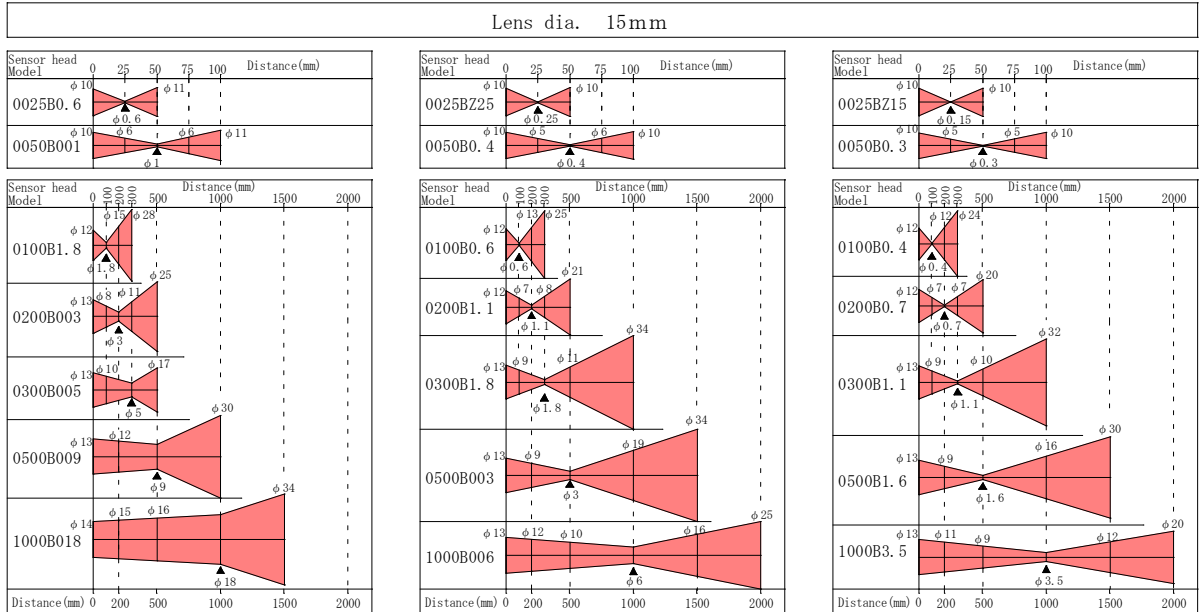
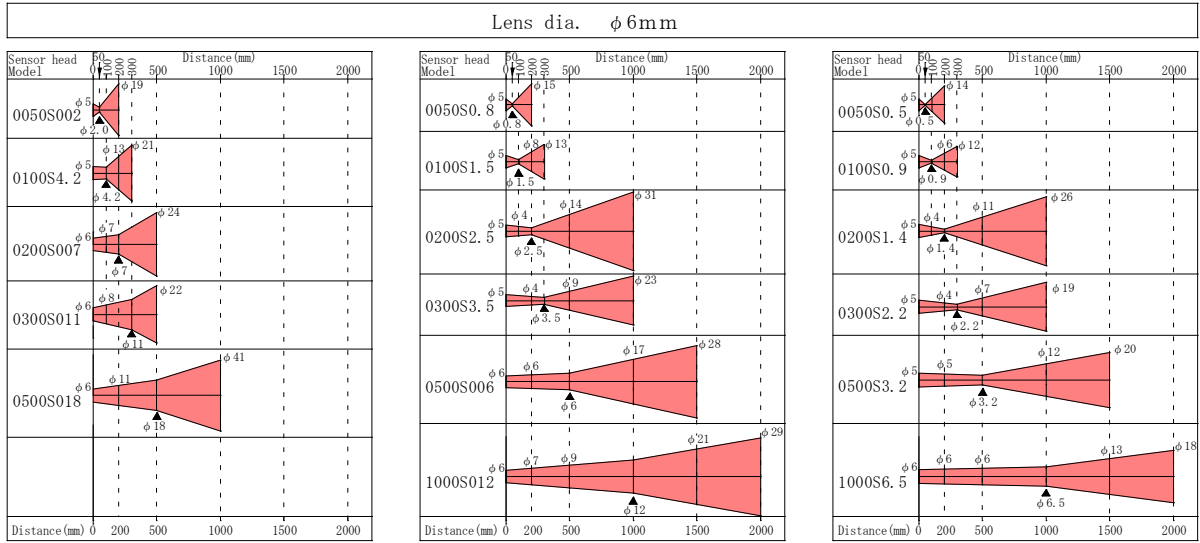
Measuring Distance and Spot Size

Note. The spot size indicates the range more than 90% of energy level.

				Temperature Converter		
				Model		
Sensor head Model	Lens dia. (mm)	Measuring distance (mm)	Target Size (mm)	ANN0500	ANN0600	ANN0700
				PNN0220	PNN0300	PNN0400
				TNN0090	TNN0160	TNN0220
0025BZ15	φ 15	25	φ 0.15	×	×	○
0025BZ25			φ 0.25	×	○	○
0025B0.6			φ 0.6	○	○	○
0050B0.3	φ 15	50	φ 0.3	×	×	○
0050B0.4			φ 0.4	×	○	○
0050B001			φ 1	○	○	○
0050S0.5	φ 6		φ 0.5	×	×	○
0050S0.8			φ 0.8	×	○	○
0050S002			φ 2	○	○	○
0100B0.4	φ 15	100	φ 0.4	×	×	○
0100B0.6			φ 0.6	×	○	○
0100B1.8			φ 1.8	○	○	○
0100S0.9	φ 6		φ 0.9	×	×	○
0100S1.5			φ 1.5	×	○	○
0100S4.2			φ 4.2	○	○	○
0200B0.7	φ 15	200	φ 0.7	×	×	○
0200B1.1			φ 1.1	×	○	○
0200B003			φ 3	○	○	○
0200S1.4	φ 6		φ 1.4	×	×	○
0200S2.5			φ 2.5	×	○	○
0200S007			φ 7	○	○	○
0300B1.1	φ 15	300	φ 1.1	×	×	○
0300B1.8			φ 1.8	×	○	○
0300B005			φ 5	○	○	○
0300S2.2	φ 6		φ 2.2	×	×	○
0300S3.5			φ 3.5	×	○	○
0300S011			φ 11	○	○	○
0500B1.6	φ 15	500	φ 1.6	×	×	○
0500B003			φ 3	×	○	○
0500B009			φ 9	○	○	○
0500S3.2	φ 6		φ 3.2	×	×	○
0500S006			φ 6	×	○	○
0500S018			φ 18	○	○	○
1000B3.5	φ 15	1000	φ 3.5	×	×	○
1000B006			φ 6	×	○	○
1000B018			φ 18	○	○	○
1000S6.5	φ 6		φ 6.5	×	×	○
1000S012			φ 12	×	○	○

Light path

Note. ▲Mark indicates the target size of standard measuring distance.



Caution

The laser aim shows near the center of the target.

Also, due to the characteristics of the laser used, the shape of the aiming beam does not became a perfect circle.

Ordering Information

Classification	Model	Basic specification		Optional specification	Contents	
FL					Fiber type thermometer	
Series	HX				Converter + Sensor head	
Standard/Special		-			Standard	
		S			Special	
Detecting element		A			Si	
		P			InGaAs1.6 μm	
		R			InGaAs2.5 μm Electrical cooling	
		T			InGaAs2.5 μm Non cooling	
		W			Two colors Si/InGaAs1.6 μm	
		V			Two colors InGaAs1.6/2.5 μm	
Purpose/Characteristic		N			For general	
		U			0.1ms Ultra high speed reply	
Standards		N			Nothing	
		E			CE	
Temperature			0500		The lowest temperature	
Sight Type			-			LED (standard)
			L			Laser
Measuring distance			1000		mm	
Head shape and Lens dia.				S		φ 6 Lens head
				B		φ 15 Lens head
Target Size				1.5		φ 1.5 mm
				010		φ 10 mm
				Z25		φ 0.25 mm
Cable length				-00		2m(Standard)
				-05		5m
				-10		10m
				-Z5		0.5m
						Z:0 A:1 B:2 C:3 D:4
						E:5 F:6 G:7 H:8 N:9
Analog output					0	0~1V
					5	0~5V(Built in resistance)
					1	0~10V(Built in resistance)
				-****		4-digit In the case of custom specification only

Troubleshooting

1) Temperature can not be measured			
①Can not be a unstable display value measurement	Can not be read fast variation of the measured value →Using the function "peak hold" or "sample hold"		
	Difficult temperature of the measurement object work to read is unstable and value →Slow down the setting time of "smoothing"		
	There is an influence of sunlight, heat source and other disturbance light →Cover to keep out light around		
	Move the value in the change in the ambient temperature and humidity →Kept constant as possible the temperature and humidity of the measurement environment		
	Are affected by noise from the outside →put a ferrite core to the cable →Metal covered with such as the (shielded)		
②Is not a value that display value predicted	Settings such as emissivity, does not match the measured work • The lack of fit narrowing and adjustment such as emissivity →Use such as black body paint, thermocouple, and the re-adjustment • Contact insufficient of thermocouple →Mounting review of thermocouple Try to change such as to embedding		
	You are not using a IR thermometer for your intended use →For quartz over, for metal, I stocks IR thermometer etc. Please contact us		
③Does not change the displayed value	Is not suitable in the direction you want to measure →To confirm the changes in the measured values in a separate heat source, for example, blocked by hand		
	Do not communicate. Does not light sighting light →Turn off the power supply, please contact us		
2) Does not work sighting light			
①The bad shape of the sighting light, Well invisible ,Is dark sighting light	Measured distance does not match →Adjust the distance to the distance of the specification Measured distance is too long. →To use the laser pointer (sold separately)		
	sighting light ON · OFF setting →Do not set to "OFF" setting Supply voltage (4.7 ~ 27V) or are supplied to →Check the power supply Do not communicate. Does not light sighting light →Turn off the power supply, please contact us		
3) cannot communicate			
①Occur frequently communication error ②Stops measured by the communication error	Is carelessly long communication cable →To fit the the cable length, and change the speed (The reference to the table below)		
	Cable length	Recommended communication maximum speed	interval
	10m below	115.2kbps	About 5ms
	20m	57.6kbps	About 10ms
	30m	57.6kbps	About 10ms
50m	38.4kbps	About 15ms	
• We display setter (TMCX series) If you are using a can, of TMCX side Since there is a description of the communication conditions in the instruction manual, please read them as well			
You are using in noisy environments →Try to drop the communication speed			
①Can not communicate	Not communication conditions are in agreement →Match the conditions		
	Wiring there is a mistake in Or I have broken →Turn off the power, to redo the wiring again		
	Do not communicate. Does not light sighting light →Turn off the power supply, please contact us		

4) Does not come out analog output	
①Does not change settings of the analog output	Output value is unstable →Line of analog output is broken
②After setting, it does not behave like analog output is thought	Consider the possibility of misconfiguration →Check whether the following setting is not wrong ①Output setting of "analog output type 1" ②Range setting of "analog output 1H", "analog output 1L"
①The output does not Change like fixation	Analog output voltage is out similar to the power supply voltage. →There is a possibility of mistake in setting Check whether the following setting is not wrong ①Output setting of "analog output type 1"
	Analog Output is 0V • Measured temperature is lower than the lower limit of the measurement range of the thermometer →If you are using is not within the considered measurement range thermometer (or does not match the application), please contact us →Check whether the following setting is not wrong • Wiring there is a mistake in Or I have broken →Turn off the power, to redo the wiring again (refer to P4)
	"Error on output" has occurred • I will be thinking is "error factor" of the following ①Reduction of the power supply voltage ②Supply voltage drop of the internal thermometer →In this case, if you are "also does not communicate, aiming light is not lit" is also combined, have come as a symptom, turn off the power, please contact us
	Do not communicate. Does not light sighting light →Turn off the power supply, please contact us
5) Does not come out Alarm output	
①After setting, it does not behave like analog output is thought	The condition of operating the alarm is not suitable • Possibility of misconfiguration →Review the condition value, such as "alarm mode 1" and "alarm 1H", "alarm 1L", to re-set.
②The output does not Change like fixation	The condition of operating the alarm is not suitable →Possibility of misconfiguration →Review the condition value, to re-set
	Power is not supplied to the alarm →And supplies power to the alarm
	I have the wrong connection Or line is broken →Turn off the power, to redo the wiring again (refer to P4)
	Do not communicate. Does not light sighting light →Turn off the power supply, please contact us

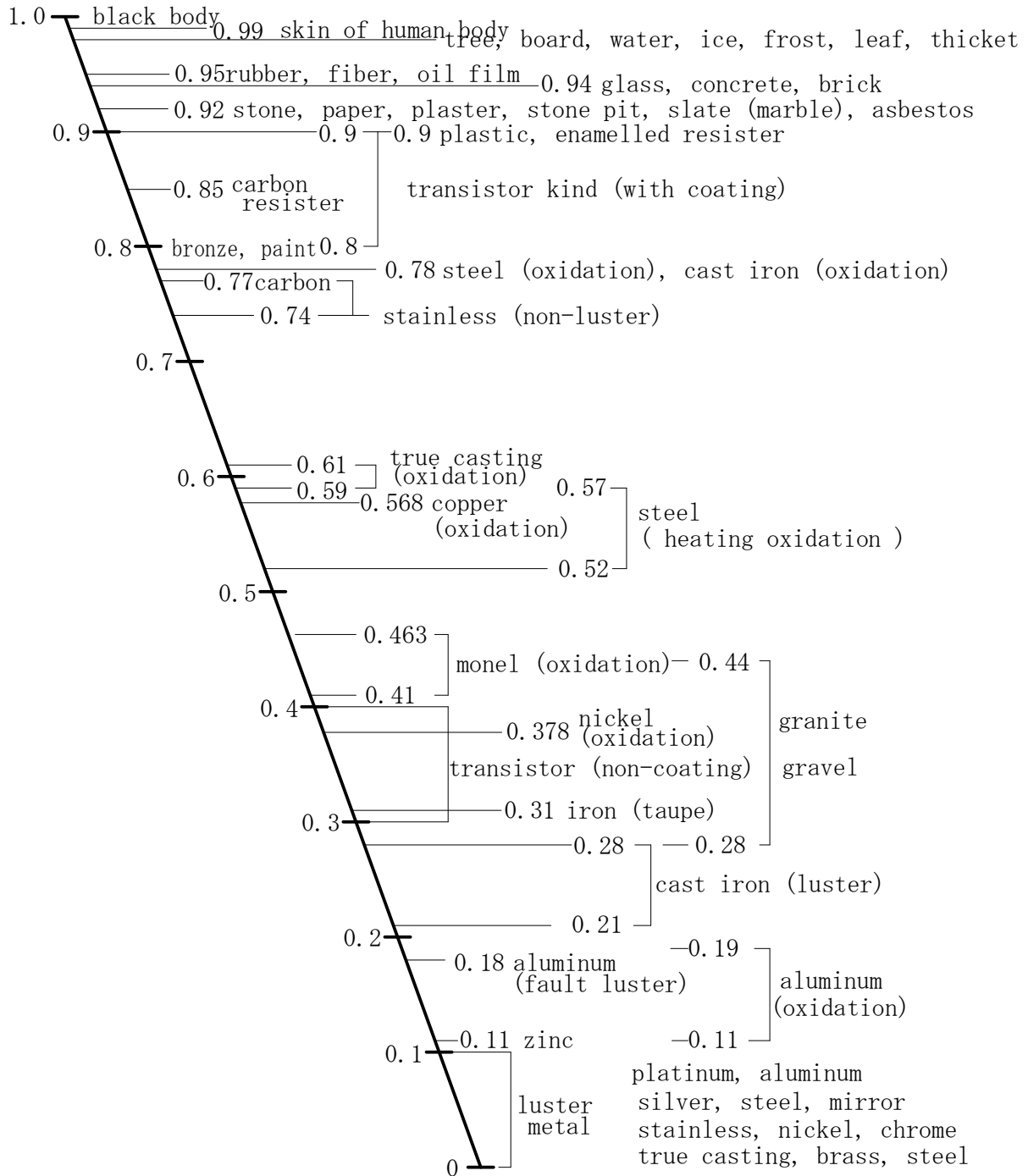
○ If you contact us, the item you want me to check in advance will be following.

①Model name of the thermometer	It is described in the instrument. "FLHX - **", etc.
②Serial number	It is described in the instrument. "V - **", etc.
③Situation you are using	Power supply voltage ** V, analog output (voltage) in use, etc.
④Specific symptoms	After the power is turned on, is not stable value even after the lapse of a few minutes (4 ~ 5 °C), etc.
⑤Measured (and temperature)	Such as "300 °C of metal surface"
⑥Measurement environment, information of installation, etc.	environment (** °C, **%), Indoor, is made to work with equipment etc.
⑦Occurrence timing and frequency of symptoms	Example: from last month, caused 10 several times in the morning, etc.

When you contact us, it is smoothly question for addressing. I wish your best regards cooperation.

Emissivity Reference Data

The typical emissivity near the normal temperature



(Note)

When measuring the temperature of the work, emissivity is important. Therefore, please measure the temperature of work by any other method (thermocouple thermometer, etc), or spread paint of which emissivity is known on the work and confirm emissivity by actual measurement.

Warranties and After sale Service

1. About the warranty period

The warranty period is 1 year from a purchase day. In the warranty period, when breaking down in the normal busy-condition which followed an instruction manual, we repair for free.

It is repaired after returned to us because of the necessary of proofreading.

2. Even if the warranty period, it is paid in the following cases.

(1) The mistake of use and the fault which was received by the other equipment.

The damage of the repair, reconstruction and the cleaning being done not by us.

(2) The trouble and damage by the moving, shipping or the fall after buying.

(3) The trouble and damage by the fire, the earthquake, the storm, the thunder, the other disasters, the pollution, the damage from salt water or abnormal-voltage, etc.

(4) The exchange by the consumption of the attachment.

(5) This warranty and the after-sale service are valid only in Japan-domestic.

3. About the repair outside of the warranty period

For repair outside of the warranty period please consult the buying agent or us.

Depending on the request of customer, it is repaired by being paid in case of maintaining with the repair.

4. When the consultation, inform us the following :

(1) Model (2) Serial No. (3) Condition of the trouble (4) Purchase date

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